

Inspections Framework

Guideline for reducing lifecycle defects

Agenda

☐ Overview

- What is it ?
- What's the benefit ?
- Inspections vs. Walkthroughs

☐ Vision

☐ Guidelines for Usage

☐ Standard Templates

☐ Standard Management Reporting

☐ Standard Process

☐ Rollout Plan



Overview of Inspections

Definition:

Structured Peer Review of Lifecycle artifacts (Requirements, External Design, Internal Design, Test Plans..) to uncover potential defects.

Purpose:

These peer reviews use formal meeting facilitation techniques to provide a best practice approach to identify and address critical defects that could impact downstream SDLC phases and ultimately production code.

Benefits

❑ Improved quality of production code

- Fewer production issues
- Reduced Support Costs
- Improved productivity
- Improved client satisfaction

❑ Improved SDLC artifacts

- Better Requirements
- Better Documentation
- Better Test Planning

❑ Reduced cycle time for SDLC phases

- Coding should require less time with better up-front analysis
- Testing should require less time with few defects

Defects can
cost upwards
of \$1500 in
production !

Inspections vs. Walkthroughs

- ❑ **There are two types of Peer Reviews each serving different purposes:**
 - Walkthroughs are an informal review used to confirm the understanding of the producer and validate the approach being taken.
 - Inspections are a formal review used to verify that the artifact complies with the standard of excellence. In a life cycle activity, the inspection is the exit criteria or gate that concludes the activity.

Goals for Inspections

- ❑ Show a steady trend of improvement in the quality of inspections by measuring the “maturity” of the inspections process throughout the development teams.
- ❑ Track the penetration of inspection usage throughout the development teams.
- ❑ Provide standard metrics for measuring the overall quality of our product deliverables providing insights on further improving our People, Process and Technologies.
- ❑ Reduce the overall costs of testing by reducing the amount of rework associated with testing defects.
- ❑ Improve the overall quality of documentation associated with the development of requirements and documentation within our Product Development Lifecycle.

Additional Goals for Inspections

- ❑ Actively engage business owners and project stakeholders in the inspection process thus improving penetration beyond the development areas.
- ❑ Institute management reporting to provide ROI insights on Inspections and to track adherence to best practices.
- ❑ Establish a forum for moderators to ensure consistency is applied to inspections and to nurture ideas for continually improving the inspections process.
- ❑ Appoint QA to champion Inspections Process

Guidelines for Using Inspections

Guidelines for Using Inspections								
	Artifacts							
Project Type	Risk	Requirements	External Design	Internal Design	Interface Design	File/DB Design	Test Plans	Regression Test Plan
Project: (New System)	H	x	x	x	x	x	x	x
Project: (New Technology)	H		x	x			x	
Project: (Large)	H	x	x	x	x	x	x	
Project: (Medium)	M	x	x	x			x	
Project: (Small)	M	x	x	x			x	
Project: (Multiple Teams)	H	x	x	x	x	x	x	x
Enhancement: (Very Small)	L			x				
Enhancement: (Single Team)	L	x		x				
Enhancement: (Multiple Teams)	M	x	x	x			x	
Interface Changes	M	x			x		x	
Data Store (DB / File) Changes	M	x				x	x	
Maintenance: (Single Module)	L							
Maintenance: (Multiple Modules)	L		x	x				

Inspection Planning Checklist

- ☐ Inspection Meeting Notice
- ☐ Risk Assessment
- ☐ Defect Legend
- ☐ Defect Log
- ☐ Defect Summary
- ☐ Defect ROI
- ☐ Project Summary
- ☐ Exit Checklist

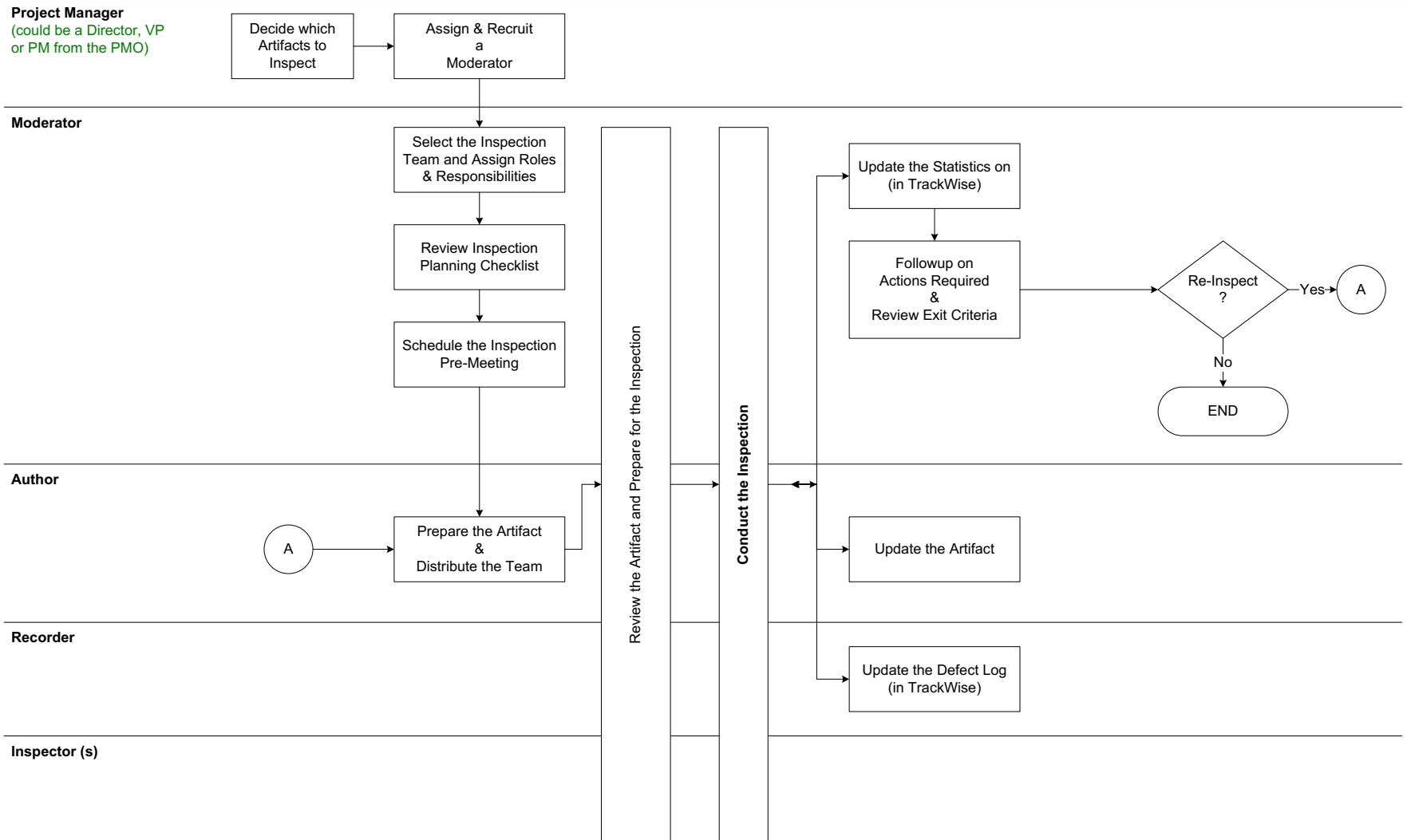
Management Reports

- ☐ Product Summary
- ☐ Team/Release Summary

Inspections Product Summary

Project	Team	Hours Spent	Inspection Costs	# of Major Defects	Costs Avoided	ROI
APO Upgrade	App Dev	20	\$1,400	15	\$22,500	\$21,100
Mfg Rollout (Wave 1)	App Dev	35	\$2,450	5	\$7,500	\$5,050
			\$0		\$0	\$0
			\$0		\$0	\$0
			\$0		\$0	\$0
			\$0		\$0	\$0
			\$0		\$0	\$0
			\$0		\$0	\$0

Process Overview

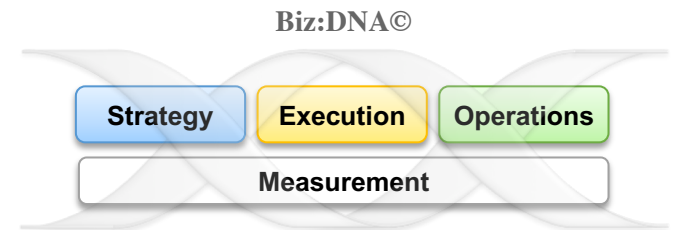


Next Steps for Moving Forward

- ☐ Project Teams are expected to begin using the process and templates going forward.
- ☐ Completed Inspections templates will be kept in each project folder in our document repositories.
- ☐ Inspection Defects will be tracked in Issue Management tool to provide centralized tracking & reporting of all defects related to inspections, walkthroughs and test case.
- ☐ An additional folder will be created to hold all Summary Sheets at the team and product level to capture all of the Inspections Related templates, processes and communications.
- ☐ IT Leaders will promote the process and provide guidance for staff.
- ☐ Incorporate Inspections into Project & Quality Plans going forward.
- ☐ Incorporate Inspections into Annual Objectives

Thank You

Measure what matters
Create and deliver the right value
Build and nurture a winning culture
Proactively manage change & complexity



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